ESERCIZIO su ANELLI di POLINOMI

$R = \mathbb{Q}[x]$

$p(x) = 2x^3 + x^2 + x - 1$

$q(x) = 2x^2 - 7x + 3$

① Trovare $MCD(p(x), q(x))$ usando algoritmo di Euclide

② $I = (2x - 1)$ ideale principale

Descrivere I

③ Stabilire se il MCD appartiene a I

④ Scrivere, se $\exists$, un omomorfismo di anelli $\varphi : \mathbb{R} \to \mathbb{Q}$ tale che $\ker(\varphi) = I$

①

$$
\begin{array}{l|l}
2x^3 + x^2 + x - 1 & 2x^2 - 7x + 3 \\
2x^3 - 7x^2 + 3x & \\
\hline
8x^2 - 2x - 1 & x + 4 \\
8x^2 - 28x + 12 & \\
\hline
26x - 13 &
\end{array}
$$

$26\left(x - \frac{1}{2}\right)$

$$
\begin{array}{l|l}
2x^2-7x+3 & x-\frac{1}{2} \\
\cline{2-2}
2x^2-x & 2x-6 \\
\cline{1-1}
\quad -6x+3 & \\
\quad -6x+3 & \\
\cline{1-1}
\qquad 0 &
\end{array}
$$

$MCD = (26x-13)$ o

$\left(x-\frac{1}{2}\right)$

(2) $I = (2x-1)$

$I = (a) = \{ a \cdot r : r \in R \}$

$I = \{ \underbrace{(2x-1) \cdot p(x)}_{(x-\frac{1}{2})} : p(x) \in R \}$

(3) • $x - \frac{1}{2} \in I$?

sì perché $(2x-1) \cdot \frac{1}{2} = \left(x - \frac{1}{2}\right)$

• $x^2 - \frac{1}{4} \in I$? $\left(x-\frac{1}{2}\right)\left(x+\frac{1}{2}\right)$

• $x^2 + x - \frac{3}{4} \in I$? sì

perché $\left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right) - \left(\frac{3}{4}\right) = 0$

TEO DEL RESTO

$$\boxed{p(x) \text{ è divisibile per } x-a \iff p(a)=0}$$

(2) $I = \{(2x-1)\cdot p(x) : p(x) \in R\} =$

= polinomi divisibili per $(2x-1)$

$= \{p(x) : p(\tfrac{1}{2}) = 0\}$

(4) $\varphi : R \longrightarrow \mathbb{Q}$

$p(x) \longmapsto p\left(\frac{1}{2}\right)$

$\ker \varphi = \{p(x) : \varphi(p(x)) = 0\} = I$

Verifichiamo che è OMO di ANELLI

$$\varphi\left(p_1(x)+p_2(x)\right) = \varphi\left(p_1(x)\right)+\varphi\left(p_2(x)\right)$$

$$p_1\left(\tfrac{1}{2}\right)+p_2\left(\tfrac{1}{2}\right) = p_1\left(\tfrac{1}{2}\right)+p_2\left(\tfrac{1}{2}\right)$$

$$\varphi\left(p_1(x)\cdot p_2(x)\right) = \varphi\left(p_1(x)\right)\cdot\varphi\left(p_2(x)\right)$$

$$p_1\left(\tfrac{1}{2}\right)\cdot p_2\left(\tfrac{1}{2}\right) = p_1\left(\tfrac{1}{2}\right)\cdot p_2\left(\tfrac{1}{2}\right)$$

5 Applichiamo TEO FOND

$$R/\ker\varphi \simeq \operatorname{Im}\varphi = \mathbb{Q}$$

$\ker\varphi = I = (2x-1)$ NON è INI

$\operatorname{Im}\varphi = \mathbb{Q}$ φ è SURIETTIVA perché per esempio

$$\varphi(a) = a \qquad a \in \mathbb{Q}$$

COSTANTE

$$\mathbb{Q}[x] \big/ (2x-1) \simeq \mathbb{Q} \quad \text{è CAMPO}$$

ANELLO QUOZIENTE

$\mathbb{Q}[x]$ anello dei poli

è un campo? NO

perché per esempio $x+1$ $\quad 2x$

x^2-3

$\frac{1}{x+1}$ non e' un polinomio

SCHEDA 3

$f\ (\mathbb{Z},+) \to (\mathbb{Z},+)$ $\qquad f(x) = 2x$ $\quad$ Sì

$g: (\mathbb{Z}^*,\cdot) \to (\mathbb{Z},+)$ $\qquad g(x) = 2x$

$$g(x_1 \cdot x_2) = g(x_1) + g(x_2)$$

$$\begin{array}{cccc} \| & \| & \| & \| \\ 1 & 1 & 2 & 2 \end{array}$$

$$g(1) = 2 \neq 2+2 = 4$$

$h: (\mathbb{R}^*,\cdot) \to (\mathbb{R}^*,\cdot)$

$$2xy \neq (2x)(2y)$$

ESEMPIO $x = y = 1$

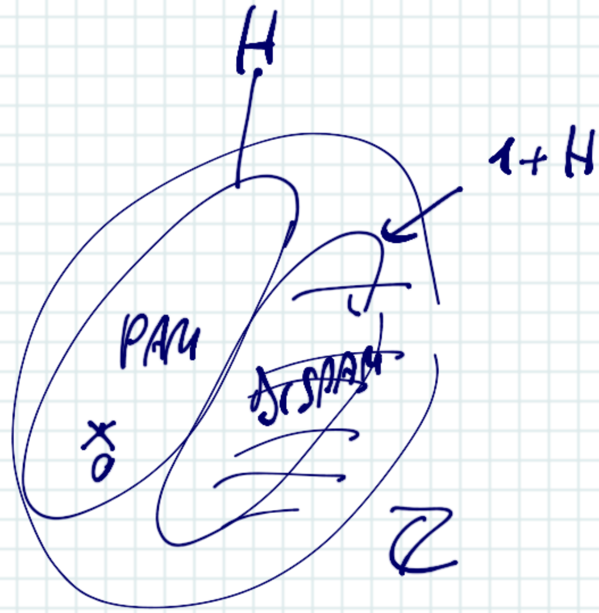
$H: 2\mathbb{Z} \leq \mathbb{Z},+$

classi laterali

$$n_0 + 2\mathbb{Z} = \{ n_0 + 2k \}$$

$0 + 2\mathbb{Z} = 2\mathbb{Z}$ PARI

$1 + 2\mathbb{Z}$ DISPARI



$$\mathbb{Z}/2\mathbb{Z} = \mathbb{Z}/H = \mathbb{Z}_2$$

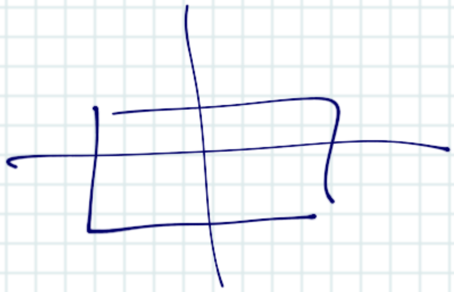
Può essere che $f^{-1}(H) = \emptyset$? NO

$f : G \to G'$ $\qquad f^{-1}(a+H) = \emptyset$?

classe laterale

$$f : \mathbb{Z} \longrightarrow \mathbb{Z}$$
$$n \longmapsto 8n$$

②



id
(12) (34)
(13) (24)
(14) (23)

V

	id	α (12)(34)	β (13)(24)	γ (14)(23)
id	id	α	β	γ
α (12)(34)	α	id	γ	β
β (13)(24)	β	γ	id	α
γ (14)(23)	γ	β	α	id

$\langle id \rangle = \{ id \}$

$\langle \alpha \rangle = \{ \alpha, \alpha^2 = id \}$ ORDINE 2

$\langle \alpha, \beta \rangle = \{ \alpha, \beta, \alpha^2 = id, \beta^2 = id, \alpha\beta = \gamma \} = G$

	α	β	id	γ
α	id	γ		
β				

	α	id
α	id	α
id	α	id

$$H = \langle (13), (24) \rangle = \{ (13), (24), id, (13)(24) \}$$

	α (13)	β (24)	id	γ $(13)(24)$
(13)	id	$(13)(24)$	.	.
(24)	$(13)(24)$	id	.	.
id	(13)	(24)	id	$(13)(24)$
$(13)(24)$	(24)	(13)	.	.

GL_2

$$H = \left\langle \underset{\overset{\shortparallel}{A}}{\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}} \right\rangle = \left\{ \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}^n \right\}$$

A^2, A^3, A^4

$$= \left\{ \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}, \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \right\}$$

$A \qquad \underset{-I}{\overset{A^2}{\shortparallel}} \qquad \underset{-A}{\overset{A^3}{\shortparallel}} \qquad \underset{I}{\overset{A^4}{\shortparallel}}$

$$A^5 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \cdot A = A$$

	A	$-I$	$\overset{A^3}{\shortparallel}$ $-A$	I
A	$-I$	$-A$	I	A
$-I$				
$-A$				
I				

$(\mathbb{C}^*, \cdot)$ $\qquad \langle i \rangle = \{ i, i^2, i^3, i \dots \}$

$$= \{ i, -1, -i, 1 \} = \langle -i \rangle$$

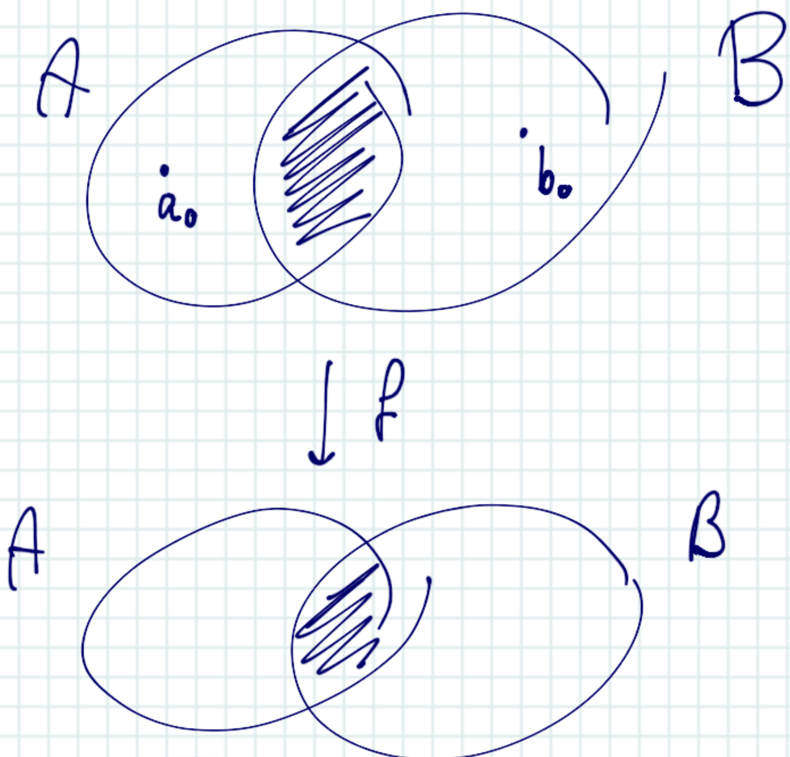
	i	-1	$-i$	1

$$\varphi : A \to i$$

generatore nel generatore

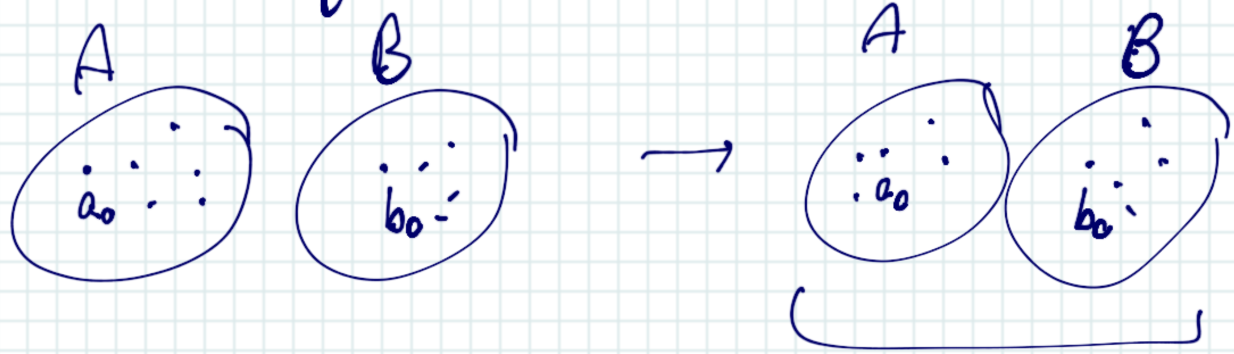
$A \to i$	$A \to -i$?
$A^2 \to i^2$	$A^2 \to -1$
$A^3 \to i^3$	$A^3 \to i$
$A^4 \to i^4$	$A^4 \to 1$

ESERCIZIO 4 SCHEDA 1



$$\boxed{f(e) = \begin{cases} b_0 & \text{se } e \in A \\ a_0 & \text{se } e \in B \end{cases}}$$

b) A e B disgiunti



$\mathrm{Im} f = \{a_0, b_0\}$ è SURIETTIVA se

$A = \{a_0\}$ e $B = \{b_0\}$

è INIETTIVA se $A = \{a_0\}$ e $B = \{b_0\}$